

UMass Math Festival

Seven Bridges of Königsberg

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Introduction

Today we will work on a classical problem that goes back to 1735. The city of Königsberg (now in Russia) was set on both sides of the river Pregel, and seven bridges crossed the river, as you can see below. According to lore, the citizens of Königsberg spent their Sundays walking in the city, and they came up with a game whose goal was to devise a way to walk around the city crossing every bridge exactly once.

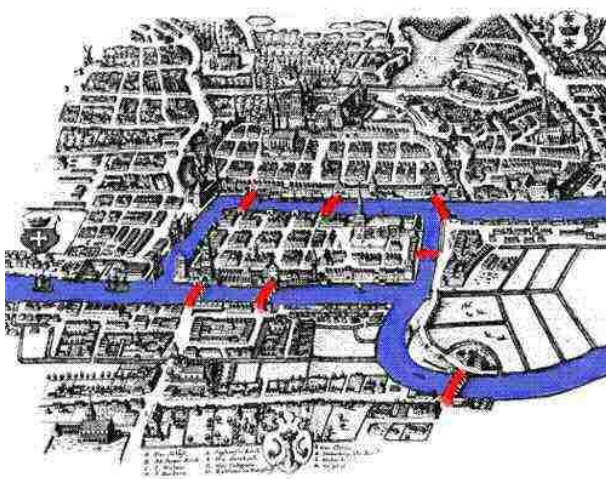


Figure 1: Seven Bridges of Königsberg

In 1736, the mayor asked Euler to solve the problem.

”. . . Thus you see, most noble Sir, how this type of solution bears little relationship to mathematics, and I do not understand why you expect a mathematician to produce it, rather than anyone else, for the solution is based on reason alone, and its discovery does not depend on any mathematical principle. Because of this, I do not know why even questions which bear so little rela-

tionship to mathematics are solved more quickly by mathematicians than by others.”

On the other hand, he wrote to a friend:

”This question is so banal, but seemed to me worthy of attention in that [neither] geometry, nor algebra, nor even the art of counting was sufficient to solve it.”

His second observation was right: the solution gave birth to a new area of mathematics, now known as graph theory.

Euler started working on it and eventually solved it. Today we will follow in his footsteps.

We can first model the problem. To do that we will use graphs. A graph consists of vertices and edges. Since the Königsberg problem is only about bridges and imposes no condition on the mainland part of the walk, we will model bridges as edges of the graph and land as vertices. You will find Euler’s original drawing in Figure 2, naming the land A , B , C , D ; these will be the vertices of our graph. Draw that graph yourself before going on.

We next define an Euler cycle to be a walk in this graph which starts and ends at the same point and crosses every edge (bridge) exactly once. Observe that we can use this definition for any graph, not necessarily the Königsberg one. (Who knows, maybe there is a city somewhere with rivers and bridges such that there is an Euler cycle.)

A few more definitions: the number of edges connected to a vertex is called the degree of the vertex (this is the same as asking how many bridges are connected to the land represented by that vertex).

And finally, we say a graph is complete if there exists exactly one edge connecting each pair of vertices.

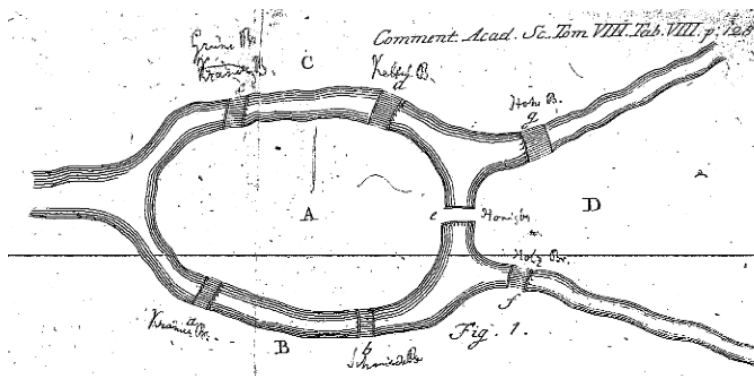


Figure 2: Euler’s original figure

1. To warm up, let us consider a graph with two vertices and one edge connecting them. Does it have an Euler cycle?
2. How about a graph with two vertices and more edges connecting them? Do they have an Euler cycle?
3. Another direction for warm-up. Recall that we call a connected graph a tree if it contains no cycles. Trees are among the most commonly used graphs. Can they have an Euler cycle?

4. We can now turn our attention to the complete opposite of trees, complete graphs: Recall, these are graphs which have a unique edge connecting every pair of vertices. Do they have Euler cycles? (Start from K_2 .)

5. What are the degrees of vertices in all of these examples?

6. Guessing part: Looking at degrees of examples and non-examples what can you say?

7. How about the Königsberg bridges? Do they have an Euler cycle?

Euler then thought of a variation of this problem: well maybe we should give up insisting this walk ends up where we begin. Rather we allow walks which use every edge once but do not necessarily start/end at the same vertex. These are called Euler paths and we can now study the question of Euler paths as well.

8. Do we really need to have only even degree vertices if we accept Euler paths? Give 5 different graph examples that have an Euler path but also have some odd-degree vertices.

9. In the examples you find, calculate the degrees of the vertices. Is there any pattern?

10. Guessing part: Looking at degrees of examples and non-examples what can you say about what we need in order to have an Euler path?

11. How about the Königsberg bridges? Do they have an Euler path?
12. What if the city builds some extra bridge(s) just to reach the goal of having an Euler path/cycle? Can it be done? What is the minimal number of bridges needed for an Euler path, and for an Euler cycle?
13. When do we have an Euler path in complete graphs?

You are welcome to use and adapt this worksheet in your own classroom; please keep the attribution. The original is available at <https://yasinkaracan.github.io>.